Chapter 1

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Exercise 1. A nonempty family of sets $\mathcal{R} \in \mathcal{P}(X)$ is called a **ring** if it is closed under finite unions and differences (i.e., if $E_1, \ldots, E_n \in \mathcal{R}$, then $\cup_1^n E_j \in \mathcal{R}$, and if $E, F \in \mathcal{R}$, then $E \setminus F \in \mathcal{R}$). A ring that is closed under countable unions is called a σ**-ring**.

a. Rings (resp. σ-rings) are closed under finite (resp. countable) intersections.

b. If $\mathcal{R}$ is a ring (resp. σ-ring), then $\mathcal{R}$ is an algebra (resp. σ-algebra) iff $X \in \mathcal{R}$.

c. If $\mathcal{R}$ is a σ-ring, then $\{E \subset X : E \in \mathcal{R} \text{ or } E^c \in \mathcal{R}\}$ is a σ-algebra.

d. If $\mathcal{R}$ is a σ-ring, then $\{E \subset X : E \cap F \in \mathcal{R} \text{ for all } F \in \mathcal{R}\}$ is a σ-algebra.

Proof.

a. If $\mathcal{R}$ is a σ-ring and $\{E_n\}_{n\in\mathbb{N}} \subset \mathcal{R}$, then

$$\bigcap_{n\in\mathbb{N}} E_n = \left(\bigcup_{n\in\mathbb{N}} E_n\right) \setminus \left(\bigcup \{E_a \setminus E_b : a, b \in \mathbb{N}\}\right) \tag{1}$$

Considering the left and right components of the difference in the right hand side of (1), the left is clearly a countable union of members of $\mathcal{R}$, and the right can be seen to be a countable union by identifying each $E_a \setminus E_b$ with $\frac{a}{b}$. Thus the right hand side of (1) is a difference of countable unions of members of $\mathcal{R}$, so it is in $\mathcal{R}$.

If $\mathcal{R}$ is a ring, the argument above applies by making $\{E_n\}$ a finite set.

b. In either case, if $\mathcal{R}$ is a ring (resp. σ-ring) and an algebra (resp. σ-algebra), then given $E \in \mathcal{R}$, $X = E \cup E^c \in \mathcal{R}$.

Conversely, if $\mathcal{R}$ is a ring (resp. σ-ring) and $X \in \mathcal{R}$, then given $E \in \mathcal{R}$, $E^c = X \setminus E \in \mathcal{R}$, so $\mathcal{R}$ is closed under complements. $\mathcal{R}$ is already closed under finite (resp. countable) unions, so $\mathcal{R}$ is an algebra.

c. Let $A = \{E \subset X : E \in \mathcal{R} \text{ or } E^c \in \mathcal{R}\}$. Given $E \in A$, by definition $E^c \in A$ since $(E^c)^c = E \in A$.

Let $\{E_n\}_{n\in\mathbb{N}} \subset A$, and express this set as the union of two sets, namely

$$\{E_n\}_{n\in\mathbb{N}} = \{F_j\}_{j\in J} \cup \{G_k\}_{k\in K}$$

where $\{F_j\}_{j\in J} \subset A$ is the collection of all sets in A who are members of $\mathcal{R}$ and $\{G_k\}_{k\in K} \subset A$ is the collection of sets in A who are not themselves in $\mathcal{R}$, but whose complements are. Then

$$\begin{aligned} \bigcup_{n\in\mathbb{N}} E_n &= \bigcup_{j\in J} F_j \cup \bigcup_{k\in K} G_k \\ &= \bigcup_{j\in J} F_j \cup \left(\bigcap_{k\in K} G_k^c \right)^c \qquad (1) \end{aligned}$$

Since each F_j is in $\mathcal{R}$, $\bigcup_{j\in J} F_j \in \mathcal{R} \subset A$. Also, we know from (a) that σ-rings are closed under countable intersections, so since $G_k^c \in R$ for each G_k, we have $\bigcap_{k\in K} G_k^c \in R$. Its complement is also in R since R is closed under complementation. Thus (2) is a union of two members of R, and is therefore in $R \subset A$.

d. Let $A = \{E \subset X : E \cap F \in \mathcal{R} \text{ for all } F \in \mathcal{R}\}$. Note that $\mathcal{R} \subset A$ since $\mathcal{R}$ is closed under finite intersections. Given any $E, F \in \mathcal{R}$, we have

$$E^c \cap F = (X \setminus E) \cap F = F \setminus E$$

which is a difference in $\mathcal{R}$ and is thus in $\mathcal{R} \subset A$.

Given $\{E_j\}_{j\in\mathbb{N}} \subset A$ and arbitrary $F \in A$,

$$\left(\bigcup_{j\in\mathbb{N}} E_j \right) \cap F = \bigcup_{j\in\mathbb{N}} (E_j \cap F),$$

which is a countable union of elements in $\mathcal{R}$. Thus since the choice of F was arbitrary, $\bigcup \{E_j\} \in A$.

□

Exercise 2. Complete the proof of Proposition 1.2.

Proof. The construction of open intervals from half-open intervals and rays is standard. □

Exercise 3. Let $\mathcal{M}$ be an infinite σ-algebra.

a. $\mathcal{M}$ contains an infinite sequence of disjoint nonempty sets.

b. $\text{card}(\mathcal{M}) \geq c$.

Proof.

a. Choose a sequence $\{E_n\}_{n\in\mathbb{N}}$ of nonempty sets in $\mathcal{M}$ (this requires the axiom of choice). We then derive the sequence $\{F_n\}$ defined recursively by $F_1 = E_1$ and $F_j = E_j \setminus F_{j-1}$ when $j \geq 2$. These are disjoint by construction and are all in $\mathcal{M}$ since $\mathcal{M}$ is closed under differences.

b. Choose a sequence $\{E_j\}_1^\infty$ of disjoint nonempty sets in $\mathcal{M}$ (allowable by part (a) of this exercise). Define $f : \mathcal{P}(\mathbb{N}) \to \mathcal{M}$ by $f : S \mapsto \bigcup_{k\in S} E_k$. This is well-defined since $f(S)$ is a countable union of members of $\mathcal{M}$, so $f(S) \in \mathcal{M}$. Also, f is an injection since all the E_j are disjoint. This brings us to the conclusion that since $\operatorname{im} f \subset \mathcal{M}$,

$$\operatorname{card}\mathcal{M} \geq \operatorname{card}(\operatorname{im} f) = \operatorname{card}(\mathcal{P}(\mathbb{N})) = c.$$

□

Exercise 4. An algebra $\mathcal{A}$ is a σ-algebra iff $\mathcal{A}$ is closed under countable increasing unions (i.e., if $\{E_j\}_1^\infty \subset \mathcal{A}$ and $E_1 \subset E_2 \subset \cdots$, then $\bigcup_1^\infty E_j \in \mathcal{A}$).

Proof. If $\mathcal{A}$ is a σ-algebra, then $\mathcal{A}$ is closed under countable unions in general, so it is certainly closed under countable increasing unions in particular. On the other hand, if $\mathcal{A}$ is an algebra which is closed under countable increasing unions, then take any $\{E_j\}_1^\infty \subset \mathcal{A}$. Consider that

$$\bigcup_{j=1}^{\infty} E_j = \bigcup_{j=1}^{\infty}\bigcup_{k=1}^{j} E_k. \tag{1}$$

Clearly, for any $j \in \mathbb{N}$, $\left(\bigcup_{k=1}^{j} E_k\right) \subset \left(\bigcup_{k=1}^{j+1} E_k\right)$, and each $\bigcup_{k=1}^{j} E_k$ is a finite union of elements of the algebra $\mathcal{A}$, and thus is in $\mathcal{A}$. Then the right hand side of (1) is a countable increasing union of members of $\mathcal{A}$. Thus (1) $\in \mathcal{A}$. □

Exercise 5. If $\mathcal{M}$ is the σ-algebra generated by $\mathcal{E}$, then $\mathcal{M}$ is the union of the σ-algebras generated by $\mathcal{F}$ as $\mathcal{F}$ ranges over all countable subsets of $\mathcal{E}$.

Proof. Let F be the set of all countable subsets of $\mathcal{E}$. By Lemma 1.1, since $\mathcal{F} \subset \mathcal{E} \subset \mathcal{M}(\mathcal{E})$ for each $\mathcal{F} \in F$, $\mathcal{M}(\mathcal{F}) \subset \mathcal{M}(\mathcal{E})$. Their union, then, is also a subset of $\mathcal{E}$.

We need to show, then, that $\mathcal{M}(\mathcal{E}) \subset \bigcup_{\mathcal{F}\in F}\mathcal{M}(\mathcal{F})$. If we can show that $\bigcup_{\mathcal{F}\in F}\mathcal{M}(\mathcal{F}) = \mathcal{M}(\bigcup_{\mathcal{F}\in F}\mathcal{F})$, then we will be done because $\mathcal{E} \subset \bigcup F$, so $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\bigcup F)$. That is where we now turn our focus.

Clearly $\mathcal{M}(\bigcup F) \subset \bigcup_{\mathcal{F}\in F}\mathcal{M}(\mathcal{F})$ since $\bigcup F \subset \bigcup_{\mathcal{F}\in F}\mathcal{M}(\mathcal{F})$, so we need to prove the other direction. Given $E \in \bigcup \mathcal{M}(F)$, fix some $\mathcal{F} \in F$ for which $E \in \mathcal{M}(\mathcal{F})$. Then since $\mathcal{F} \subset \bigcup F$, $\mathcal{M}(\mathcal{F}) \subset \mathcal{M}(\bigcup F)$, so $E \in \mathcal{M}(\bigcup F)$. As observed above, this completes our proof. □

Exercise 6. Complete the proof of Theorem 1.9.

Proof. First we verify that $\overline{\mu}$ is a complete measure on $\overline{\mathcal{M}}$. Since we can write $\emptyset = \emptyset \cup \emptyset$ where $\emptyset \in \mathcal{M}, \mathcal{N}$, we have $\overline{\mu} = \mu(\emptyset) = 0$. Given a disjoint collection $\{E_j \cup F_j\}_{j \in \mathbb{N}} \subset \overline{\mathcal{M}}$ where each $E_j \in \mathcal{M}$ and each $F_j \subset N_j \in \mathcal{N}$, we have

$$\begin{aligned}
\overline{\mu}\left(\bigcup_1^\infty (E_j \cup F_j)\right) &= \overline{\mu}\left(\bigcup_1^\infty E_j \cup \bigcup_1^\infty F_j\right) \\
&= \mu\left(\bigcup_1^\infty E_j\right) \\
&= \sum_1^\infty \mu(E_j) \\
&= \sum_1^\infty \overline{\mu}(E_j \cup F_j)
\end{aligned}$$

where the second equality holds because $\bigcup_1^\infty E_j \in \mathcal{M}$ and $\bigcup_1^\infty F_j \in \mathcal{N}$ since both $\mathcal{M}$ and $\mathcal{N}$ are closed under countable unions. Of course, since $\overline{\mu}$ is a measure and its domain is complete, $\overline{\mu}$ is a complete measure.

Now we turn our attention to the uniqueness of $\overline{\mu}$ as an extension of μ over $\overline{\mathcal{M}}$. Assume that f is a measure over $\overline{\mathcal{M}}$ which extends μ. Then given $M \cup F \in \overline{\mathcal{M}}$ where $M \in \mathcal{M}$ and $F \subset N \in \mathcal{N}$, by monotonicity we have that $f(F) \leq f(N) = 0$, so $f(F) = 0$. We have

$$f(M) \leq f(M \cup F) \leq f(M) + f(F) = f(M).$$

Thus $f(M \cup F) = f(M) = \mu(M) = \overline{\mu}(M \cup F)$. □

Exercise 7. If $\mu_1, \ldots, \mu_n$ are measures on $(X, \mathcal{M})$ and $a_1, \ldots, a_n \in [0, \infty)$, then $\sum_1^n a_j \mu_j$ is a measure on $(X, \mathcal{M})$.

Proof. Clearly $\sum_1^\infty a_j \mu_j(\emptyset) = 0$. Given a disjoint collection $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$, we have

$$\begin{aligned}
\sum_{k=1}^n \left(a_k \mu_k \left(\bigcup_{j=1}^\infty E_j\right)\right) &= \sum_{k=1}^n \left(a_k \sum_{j=1}^\infty \mu_k(E_j)\right) \\
&= \sum_{k=1}^n \sum_{j=1}^\infty a_k \mu_k(E_j) \\
&= \sum_{j=1}^\infty \sum_{k=1}^n a_k \mu_k(E_j).
\end{aligned}$$

□

Exercise 8. If $(X, \mathcal{M}, \mu)$ is a measure space and $\{E_j\}_1^\infty \subset \mathcal{M}$, then $\mu(\liminf E_j) \leq \liminf \mu(E_j)$. Also, $\mu(\limsup E_j) \geq \limsup \mu(E_j)$ provided that $\mu(\bigcup_1^\infty E_j) < \infty$.

Proof. Observe that

$$\begin{aligned}
\mu(\liminf E_j) &= \mu\left(\bigcup_{k=1}^{\infty}\bigcap_{j=k}^{\infty} E_j\right) \\
&= \lim_{k\to\infty} \mu\left(\bigcap_{j=k}^{\infty} E_j\right) \\
&\leq \lim_{k\to\infty} \mu\left(E_k\right) \\
&\ldots \text{UNFINISHED} \\
&= \sup_{k\geq 1}\inf_{j\geq k} \mu(E_j) \\
&= \liminf \mu(E_j)
\end{aligned}$$

□

Exercise 9. If $(X, \mathcal{M}, \mu)$ is a measure space and $E, F \in \mathcal{M}$, then $\mu(E)+\mu(F) = \mu(E \cup F) + \mu(E \cap F)$.

Proof.

$$\begin{aligned}
\mu(E \cup F) &= \mu((E \setminus F) \cup (E \cap F) \cup (F \setminus E)) \\
&= \mu(E \setminus F) + \mu(E \cap F) + \mu(F \setminus E) \\
&= \mu(E \setminus F) + \mu(E \cap F) - \mu(E \cap F) + \mu(E \cap F) + \mu(F \setminus E) \\
&= \mu((E \setminus F) \cup (E \cap F)) - \mu(E \cap F) + \mu((E \cap F) \cup (F \setminus E)) \\
&= \mu(E) - \mu(E \cap F) + \mu(F).
\end{aligned}$$

Thus, $\mu(E \cup F) + \mu(E \cap F) = \mu(E) + \mu(F)$ as desired. □

Exercise 10. Given a measure space $(X, \mathcal{M}, \mu)$ and $E \in \mathcal{M}$, define $\mu_E(A) = \mu(A \cap E)$ for $A \in \mathcal{M}$. Then μ_E is a measure.

Proof. First off, $\mu_E(\emptyset) = \mu(\emptyset \cap E) = \mu(\emptyset) = 0$. Now suppose that we have some

$\{E_j\}_{j\in\mathbb{N}} \subset \mathcal{M}$ a disjoint collection. Then

$$\begin{aligned}
\mu_E\left(\bigcup_{j=1}^{\infty} E_j\right) &= \mu\left(\left(\bigcup_{j=1}^{\infty} E_j\right)\cap E\right)\\
&= \mu\left(\bigcup_{j=1}^{\infty} E_j \cap E\right)\\
&= \sum_{j=1}^{\infty} \mu\left(E_j \cap E\right)\\
&= \sum_{j=1}^{\infty} \mu_E\left(E_j\right).
\end{aligned}$$

$\square$

Exercise 11. A finitely additive measure μ is a measure iff it is continuous from below as in Theorem 1.8c. If $\mu(X) < \infty$, μ is a measure iff it is continuous from above as in Theorem 1.8d.

Proof. For both assertions, if μ is a measure, then clearly it is continuous from above and below so we restrict ourselves to proving the other direction in either case. Also, $\mu(\emptyset) = 0$ since μ is a finitely additive measure in both cases, so we will only prove countable additivity.

Given that μ is a finitely additive measure and it is continuous from below, consider a disjoint collection $\{E_j\}_{j\in\mathbb{N}} \subset \mathcal{M}$. Consider the sequence defined by $F_j = \bigcup_{k=1}^{j} E_k$, which is clearly an ascending chain. Then

$$\begin{aligned}
\mu\left(\bigcup_{j=1}^{\infty} E_j\right) &= \mu\left(\bigcup_{j=1}^{\infty} F_j\right)\\
&= \lim_{j\to\infty} \mu(F_j)\\
&= \lim_{j\to\infty} \mu\left(\bigcup_{k=1}^{j} E_k\right)\\
&= \lim_{j\to\infty} \sum_{k=1}^{j} \mu(E_k)\\
&= \sum_{j=1}^{\infty} \mu\left(E_j\right).
\end{aligned}$$

Similarly, assuming that $\mu(X) < \infty$ and μ is continuous from above, define

$\{F_j\}$ as above. Observe that $\{F_j^C\}$ is a descending chain, so

$$\begin{aligned}
\mu\left(\bigcup_{j=1}^{\infty} E_j\right) &= \mu\left(\bigcup_{j=1}^{\infty} F_j\right) \\
&= \mu\left(\left(\bigcap_{j=1}^{\infty} F_j^C\right)^C\right) \\
&= \mu\left(X \setminus \left(\bigcap_{j=1}^{\infty} F_j^C\right)\right) \\
&= \mu(X) - \mu\left(\bigcap_{j=1}^{\infty} F_j^C\right) \\
&= \mu(X) - \lim_{j\to\infty} \mu(F_j^C) \\
&= \mu(X) - \lim_{j\to\infty} \mu(X \setminus F_j) \\
&= \mu(X) - \lim_{j\to\infty} (\mu(X) - \mu(F_j)) \\
&= \mu(X) - \lim_{j\to\infty} \mu(X) + \lim_{j\to\infty} \mu(F_j) \\
&= \lim_{j\to\infty} \mu(F_j) \\
&= \sum_{j=1}^{\infty} \mu(E_j).
\end{aligned}$$

□

Exercise 12. Let $(X, \mathcal{M}, \mu)$ be a finite measure space.

- **a.** If $E, F \in \mathcal{M}$ and $\mu(E \bigtriangleup F) = 0$, then $\mu(E) = \mu(F)$.
- **b.** Say that $E \sim F$ if $\mu(E \bigtriangleup F) = 0$; then $\sim$ is an equivalence relation on $\mathcal{M}$.
- **c.** For $E, F \in \mathcal{M}$, define $\rho(E, F) = \mu(E \bigtriangleup F)$. Then $\rho(E, G) \leq \rho(E, F) + \rho(F, G)$, and hence ρ defines a metric on the space $\mathcal{M}/\sim$ of equivalence classes.

Proof.

- **a.** Observe that

$$0 = \mu(E \bigtriangleup F) = \mu(E \cup F) - \mu(E \cap F),$$

so

$$\begin{aligned}\mu(E \cap F) &= \mu(E \cup F) \\ &= \mu((E \setminus F) \cup F) \\ &= \mu(E \setminus F) + \mu(F) \\ &= \mu(F)\end{aligned}$$

since $E \setminus F \subset E \triangle F$, and thus $\mu(E \setminus F) = 0$ by monotonicity. Swapping E and F in the above calculation reveals that $\mu(E \cap F) = \mu(E)$ as well, so $\mu(F) = \mu(E)$.

b. Surely given $E \in \mathcal{M}$, $\mu(E \triangle E) = \mu(\emptyset) = 0$, so $E \sim E$. Given also $F \in \mathcal{M}$ such that $E \sim F$, then since $E \triangle F = F \triangle E$, $\mu(F \triangle E) = 0$ so that $F \sim E$. Finally, if we have $G \in \mathcal{M}$ such that $F \sim G$, then notice that

$$E \triangle G \subset (E \triangle F) \cup (F \triangle G).$$

The easiest way to see this is with a venn diagram (shoot me), but it can also be shown with some slightly painful set theory. From this, we use monotonicity and subadditivity (respectively) to get

$$\begin{aligned}\mu(E \triangle G) &\leq \mu((E \triangle F) \cup (F \triangle G)) \\ &\leq \mu(E \triangle F) + \mu(F \triangle G) \\ &= 0,\end{aligned}$$

so $\mu(E \triangle G) = 0$.

c. By the definition of $\sim$, $\rho(E, F) = 0$ if and only if $E \sim F$. Also, $\rho(E, F) = E \triangle F = F \triangle E = \rho(F, E)$, so the problem statement is justified in that we need only prove the triangle inequality.

The following is similar to the end of **b**, but is reproduced here without the specific choices of E, F, G used in **b** and adapted for the problem at hand. Given *any* $E, F, G \in \mathcal{M}$, we have

$$\begin{aligned}\rho(E, G) &= \mu(E \triangle G) \\ &\leq \mu((E \triangle F) \cup (F \triangle G)) \\ &\leq \mu(E \triangle F) + \mu(F \triangle G) \\ &= \rho(E, F) + \rho(F, G)\end{aligned}$$

with the second line following from monotonicity and the third from subadditivity.

$\square$

Exercise 13. Every σ-finite measure is semifinite.

Proof. Given a σ-finite measure space $(X, \mathcal{M}, \mu)$, if μ is finite, then semifiniteness is obvious. Assume, then, that $\mu(X) = \infty$, and let $E \in \mathcal{M}$ such that $\mu(E) = \infty$. Also, choose $\{E_j\}_{j\in\mathbb{N}} \subset \mathcal{M}$ such that $\mu(E_j) < \infty$ for each j and $\bigcup_1^\infty E_j = X$. Then $\bigcup_1^\infty (E \cap E_j) = E$ and by monotonicity $\mu(E \cap E_j) \leq \mu(E_j) < \infty$ for all j. It cannot be the case that each $E \cap E_j$ has measure 0 since

$$\infty = \mu\left(\bigcup_1^\infty (E \cap E_j)\right) \leq \sum_1^\infty \mu(E \cap E_j),$$

so we can choose $k \in \mathbb{N}$ such that $0 < \mu(E \cap E_k) < \infty$ as desired. □

Exercise 14. If μ is a semifinite measure and $\mu(E) = \infty$, for any $C > 0$ there exists $F \subset E$ with $C < \mu(F) < \infty$.

[NOTE FROM CONNOR: I'm pretty proud of this proof.]

Proof. Let $(X, \mathcal{M}, \mu)$ be the measure space in question. Given $E \in \mathcal{M}$ such that $\mu(E) = \infty$, define $A = \{F : F \in \mathcal{M}, F \subset E, 0 < \mu(F) < \infty\}$ and $B = \{\mu(F) : F \in A\}$. Now assume (contradicting the hypothesis) that B is (finitely) bounded from above. Then B has a least upper bound b.

Choose $F_1, F_2, \dots \in A$ such that $\{\mu(F_j)\}_{j\in\mathbb{N}}$ converges to b. This is possible because b is a limit point of B. Then define $G_j = \bigcup_{k=1}^j F_k$ for all $j \in \mathbb{N}$. $\{G_j\}$ is an ascending chain of finitely measured sets, and $G_j \subset E$ for all j, so $\bigcup_{j=1}^\infty G_j \subset E$ and $\mu\left(\bigcup_{j=1}^\infty G_j\right) = \lim_{j\to\infty} \mu(G_j)$. The sequence $\{\mu(G_j)\}$ is monotonically increasing due to monotonicity of μ, and it is bounded from above because $\mu(G_j) \leq b$ for any j, so it does converge. In fact, $\mu(F_j) \leq \mu(G_j) \leq b$ for any j, which means that $\{\mu(G_j)\}$ is sandwiched between $\{\mu(F_j)\}$ (which converges to b) and the constant sequence $\{b\}$ (which also converges to b), so $\{\mu(G_j)\}$ must also converge to b. Thus $\mu\left(\bigcup_{j=1}^\infty G_j\right) = b$.

Given this, we see that $\mu\left(E \setminus \bigcup_{j=1}^\infty G_j\right) = \infty$, so there is a set $G \subset E \setminus \bigcup_{j=1}^\infty G_j$ such that $0 < \mu(G) < \infty$. Then $H = G \cup \bigcup_{j=1}^\infty G_j$ is such that $H \subset E$, $0 < \mu(H) < \infty$, and $\mu(H) > b$. Thus $H \in A$ and $\mu(H) \in B$, which contradicts our assumption of b being a supremum of B. □

Exercise 15. Given a measure μ on $(X, \mathcal{M})$, define μ_0 on $\mathcal{M}$ by $\mu_0(E) = \sup\{\mu(F) : F \subset E \text{ and } \mu(F) < \infty\}$.

- **a.** μ_0 is a semifinite measure. It is called the *semifinite part* of μ.
- **b.** If μ is semifinite, then $\mu = \mu_0$. (Use Exercise 14.)
- **c.** There is a measure ν on $\mathcal{M}$ (in general, not unique) which assumes only the values 0 and ∞ such that $\mu = \mu_0 + \nu$.

Proof.

a. Clearly $\mu_0(\emptyset) = 0$ since the only subset of $\emptyset$ is $\emptyset$, whose measure is 0. Given a disjoint collection $\{E_j\}_{j\in\mathbb{N}} \subset \mathcal{M}$, we have

$$\begin{aligned}
\mu_0\left(\bigcup_{j=1}^\infty E_j\right) &= \sup\left\{\mu(F) : F \subset \bigcup_{j=1}^\infty E_j \text{ and } \mu(F) < \infty\right\}\\
&= \sup\left\{\sum_{k=1}^\infty \mu(F\cap E_k) : F \subset \bigcup_{j=1}^\infty E_j \text{ and } \mu(F) < \infty\right\}\\
&= \sum_{k=1}^\infty \sup\left\{\mu(F\cap E_k) : F \subset \bigcup_{j=1}^\infty E_j \text{ and } \mu(F) < \infty\right\}\\
&\text{CHECK THIS SET EQUALITY CAREFULLY}\\
&= \sum_{k=1}^\infty \sup\left\{\mu(F) : F \subset E_k \text{ and } \mu(F) < \infty\right\}\\
&= \sum_{k=1}^\infty \mu_0(E_k).
\end{aligned}$$

So μ_0 is a measure.

Given any $E \in \mathcal{M}$ such that $\mu_0(E) = \infty$, by definition we see that

$$\infty = \sup\left\{\mu(F) : F \subset E \text{ and } \mu(E) < \infty\right\},$$

so certainly there exists measurable $F \subset E$ such that $0 < \mu(F) < \infty$. Thus μ_0 is semifinite.

b. Note that if $\mu(E)$ is finite, then clearly $\mu_0(E) = \mu(E)$, so we need only check agreement on measurable sets where $\mu(E) = \infty$. Given, then, $E \in \mathcal{M}$ such that $\mu(E) = \infty$, by Exercise 14 we know that

$$\{\mu(F) : F \subset E \text{ and } \mu(F) < \infty\}$$

is unbounded from above, and thus its supremum is ∞. Therefore

$$\mu_0(E) = \infty = \mu(E).$$

c. Define ν so that given $E \in \mathcal{M}$, $\nu(E) = 0$ if $\mu(E) < \infty$ and $\nu(E) = \infty$ if $\mu(E) = \infty$. As observed in **b**, $\mu(E) = \mu_0(E)$ when $\mu(E) < \infty$, so

$$\mu_0(E) + \nu(E) = \mu_0(E) = \mu(E)$$

in that case. In case $\mu(E) = \infty$, then no matter the value of $\mu_0(E)$,

$$\mu_0(E) + \nu(E) = \infty = \mu(E).$$

$\square$

Exercise 16. Let $(X, \mathcal{M}, \mu)$ be a measure space. A set $E \subset X$ is called **locally measurable** if $E \cap A \in \mathcal{M}$ for all $A \in \mathcal{M}$ such that $\mu(A) < \infty$. Let $\widetilde{\mathcal{M}}$ be the collection of all locally measurable sets. Clearly $\mathcal{M} \subset \widetilde{\mathcal{M}}$; then if $\mathcal{M} = \widetilde{\mathcal{M}}$, then μ is called **saturated**.

a. If μ is σ-finite, then μ is saturated.

b. $\widetilde{\mathcal{M}}$ is a σ-algebra.

c. Define $\widetilde{\mu}$ on $\widetilde{\mathcal{M}}$ by $\widetilde{\mu}(E) = \mu(E)$ if $E \in \mathcal{M}$ and $\widetilde{\mu}(E) = \infty$ otherwise. Then $\widetilde{\mu}$ is a saturated measure on $\widetilde{\mathcal{M}}$, called the **saturation** of μ.

d. If μ is complete, so is $\widetilde{\mu}$.

e. Suppose that μ is semifinite. For $E \in \widetilde{\mathcal{M}}$, define $\underline{\mu}(E) = \sup \{\mu(A) : A \in \mathcal{M} \text{ and } A \subset E\}$. Then $\underline{\mu}$ is a saturated measure on $\widetilde{\mathcal{M}}$ that extends μ.

f. Let X_1, X_2 be disjoint uncountable sets, $X = X_1 \cup X_2$, and $\mathcal{M}$ the σ-algebra of countable or co-countable sets in X. Let μ_0 be counting measure on $\mathcal{P}(X_1)$, and define μ on $\mathcal{M}$ by $\mu(E) = \mu_0(E \cap X_1)$. Then μ is a measure on $\mathcal{M}$, $\widetilde{\mathcal{M}} = \mathcal{P}(X)$, and in the notation of parts (c) and (e), $\widetilde{\mu} \neq \underline{\mu}$.

Proof.

a. Let $E \subset X$ be locally measurable and let $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$ such that $\mu(E_j) < \infty$ for each j and $\bigcup_1^\infty E_j = X$. Then $E = \bigcup_1^\infty (E_j \cap E)$, and because E is locally measurable, each $E_j \cap E$ is measurable, so E is a countable union of measurable sets. Thus E is measurable.

b. Given $\{E_j\}_{j \in \mathbb{N}} \subset \widetilde{\mathcal{M}}$ and $A \in \mathcal{M}$ such that $\mu(A) < \infty$,

$$\left(\bigcup_1^\infty E_j \right) \cap A = \bigcup_1^\infty (E_j \cap A),$$

so $(\bigcup_1^\infty E_j) \cap A$ is the countable union of measurable sets and is thus measurable itself.

Also, given $E \in \widetilde{\mathcal{M}}$ and A as before,

$$E^C \cap A = A \setminus (E \cap A).$$

Since $E \cap A \in \mathcal{M}$ and $\mathcal{M}$ is closed under differences, $E^C \cap A \in \mathcal{M}$.

Thus $\widetilde{\mathcal{M}}$ is a σ-algebra.

c. First we show that $\widetilde{\mu}$ is a measure. Since $\emptyset \in \mathcal{M}$, $\widetilde{\mu}(\emptyset) = \mu(\emptyset) = 0$. Also, let $\{E_j\}_{j \in \mathbb{N}} \subset \widetilde{\mathcal{M}}$ be a disjoint collection. If $\bigcup_1^\infty E_j \in \mathcal{M}$, then since each

E_j is locally measurable, $E_k = E_k \cap \bigcup_1^\infty E_j \in \mathcal{M}$ for each k. Then

$$\begin{aligned}\widetilde{\mu}\left(\bigcup_1^\infty E_j\right) &= \mu\left(\bigcup_1^\infty E_j\right)\\ &= \sum_1^\infty \mu(E_j)\\ &= \sum_1^\infty \widetilde{\mu}(E_j).\end{aligned}$$

On the other hand, if $\bigcup_1^\infty E_j \notin \mathcal{M}$, then for some $k \in \mathbb{N}$, $E_k \notin \mathcal{M}$. Thus

$$\begin{aligned}\widetilde{\mu}\left(\bigcup_1^\infty E_j\right) &= \infty\\ &= \sum_1^\infty \widetilde{\mu}(E_j)\end{aligned}$$

since $\widetilde{\mu}(E_k) = \infty$.

Now we show that $\widetilde{\mu}$ is saturated. Let $E \in X$ locally measurable over $\widetilde{\mu}$. We want to show that $E \in \widetilde{\mathcal{M}}$. Let $A \in \mathcal{M}$ such that $\mu(A) < \infty$. Then since $A \in \widetilde{\mathcal{M}}$, $E \cap A \in \widetilde{\mathcal{M}}$, and by monotonicity

$$\widetilde{\mu}(E \cap A) < \widetilde{\mu}(A) = \mu(A) < \infty.$$

Thus since $\widetilde{\mu}(E \cap A)$ is finite, $E \cap A \in \mathcal{M}$. Therefore $E \in \widetilde{\mathcal{M}}$.

d. Let $F \subset N$ for some $N \in \widetilde{\mathcal{M}}$ where $\widetilde{\mu}(N) = 0$. Then $N \in \mathcal{M}$ since $\widetilde{\mu}(N)$ is finite. Since μ is complete, $F \in \mathcal{M} \subset \widetilde{\mathcal{M}}$. Thus $\widetilde{\mu}$ is complete.

e. First we prove that $\underline{\mu}$ is a measure. Since $\emptyset$ is the only subset of $\emptyset$ and $\mu(\emptyset) = 0$, $\underline{\mu}(\emptyset) = \sup\{0\} = 0$. Also, given a disjoint collection $\{E_j\}_{j\in\mathbb{N}} \subset$

$\widetilde{\mathcal{M}}$,

$$\begin{aligned}
\underline{\mu}\left(\bigcup_1^\infty E_j\right) &= \sup\left\{\mu(A) : A \in \mathcal{M} \text{ and } A \subset \bigcup_1^\infty E_j\right\} \\
&= \sup\left\{\mu\left(\bigcup_{k=1}^\infty A \cap E_k\right) : A \in \mathcal{M} \text{ and } A \subset \bigcup_1^\infty E_j\right\} \\
&= \sup\left\{\sum_{k=1}^\infty \mu\left(A \cap E_k\right) : A \in \mathcal{M} \text{ and } A \subset \bigcup_1^\infty E_j\right\} \\
&= \sum_{k=1}^\infty \sup\left\{\mu\left(A \cap E_k\right) : A \in \mathcal{M} \text{ and } A \subset \bigcup_1^\infty E_j\right\} \\
&= \sum_{k=1}^\infty \sup\left\{\mu(A) : A \in \mathcal{M} \text{ and } A \subset E_k\right\} \\
&= \sum_{k=1}^\infty \underline{\mu}(E_k).
\end{aligned}$$

Thus $\underline{\mu}$ is a measure.

To show that it extends μ is easy since by monotonicity, any subset of some $E \in \mathcal{M}$ has measure less than or equal to that of E, so

$$\underline{\mu}(E) = \sup\left\{\mu(A) : A \in \mathcal{M} \text{ and } A \subset E\right\} = \mu(E).$$

Now we just need to show saturation. In particular, if $E \in X$ is locally measurable over $\underline{\mu}$, then we need to show that $E \in \widetilde{\mathcal{M}}$. Thus we take $A \in \mathcal{M}$ such that $\mu(A) < \infty$ and see that since $A \in \widetilde{\mathcal{M}}$, $E \cap A \in \widetilde{\mathcal{M}}$. Then $E \cap A$ is locally measurable over μ, so $E \cap A = (E \cap A) \cap A \in \mathcal{M}$. Therefore $\underline{\mu}$ is saturated.

f. First we show that μ is a measure on $\mathcal{M}$. We see that $\mu(\emptyset) = \mu_0(\emptyset \cap X_1) = \mu_0(\emptyset) = 0$. Let $\{E_j\}_{j\in\mathbb{N}} \subset \mathcal{M}$ be disjoint. Then

$$\begin{aligned}
\mu\left(\bigcup_1^\infty E_j\right) &= \mu_0\left(\left(\bigcup_1^\infty E_j\right) \cap X_1\right) \\
&= \mu_0\left(\bigcup_1^\infty (E_j \cap X_1)\right) \\
&= \sum_1^\infty \mu_0(E_j \cap X_1) \\
&= \sum_1^\infty \mu(E_j).
\end{aligned}$$

Next we want to show that $\widetilde{\mathcal{M}} = \mathcal{P}(X)$, or in other words, that every subset of X is locally measurable. Let $E \subset X$ and $A \in \mathcal{M}$ such that $\mu(A) < \infty$. Notice that $A \cap X_1$ must be finite since $\mu_0(A \cap X_1) < \infty$. Thus A^C is uncountable because X_1 is uncountable and $A^C \cap X_1 = X_1 \backslash (A \cap X_1)$. This means A cannot be co-countable, so since $A \in \mathcal{M}$, it must be countable. Seeing this, it is obvious that $E \cap A$ is countable because $E \cap A \subset A$, so $E \cap A \in \mathcal{M}$.

Finally, to see that $\widetilde{\mu} \neq \underline{\mu}$, observe that $X_2 \notin \mathcal{M}$ because X_2 is uncountable and $X_2^C = X_1$ is uncountable. Then $\widetilde{\mu}(X_2) = \infty$ because $X^2 \notin \mathcal{M}$, but $\mu(X_2) = \mu_0(X_2 \cap X_1) = \mu_0(\emptyset) = 0$, so any subset of X_2 also has measure 0. Thus

$$\underline{\mu}(X_2) = \sup\{\mu(A) : A \in \mathcal{M} \text{ and } A \subset X_2\} = \sup\{0\} = 0.$$

We conclude that since $\widetilde{\mu}(X_2) = \infty \neq 0 = \underline{\mu}(X_2)$, $\widetilde{\mu} \neq \underline{\mu}$.

$\square$